

yelp* Negativity and Usefulness

POLI 176 - Ariel Tran





Research Question:

Are yelp reviews that are more negative more likely to be perceived as “useful”?

Why is this Important?

- Predicting usefulness of Yelp reviews (Luo 2019)
 - positive + food, negative + value
- Extremity and Negativity Biases (Minjung 2021)
 - Negative > positive
- Exploring the relationship between negativity and usefulness can guide review platforms to be more effective

yelp Dataset

University of California San Diego

3.6 (237 reviews)

Claimed • Colleges & Universities

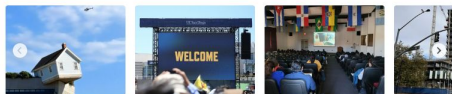
Closed 9:00 AM - 5:00 PM See hours

Write a review Add photos/videos Share Save

Do you recommend this business? Yes No Maybe

Photos & videos

See all 647 photos →



- Founded in 2004
- Crowd-sourced reviews on business
- In recent year, not so web scraping friendly

- Data from 2005 - 2013
- Official data from the company
- 10,000 reviews/rows
- Heavily right-skewed



Sanji S. Elite 25
Los Angeles, CA
9 53 135

Feb 9, 2023

It's actually WILD how I got a parking ticket this morning because I parked in a "B" spot since I had no where else to park before my midterm and I would have been late otherwise.

It's insane that they're building more housing when they don't even have enough parking structures for their students who currently go there! Why are we even paying for parking spots as students if there aren't even enough S spots for us to park in???? We're forced to park in staff spots and then they cite us a \$65 ticket?!?! As if the tuition isn't enough, I now have to pay for this inflated parking ticket for trying to go to the class I pay for here at this overpriced institution.



Helpful 1



Thanks 2



Love this 1



Oh no 0

My Process

Hand Coding

100 randomly
selected reviews

2 coders

1 = negative

0 = positive

ML Models

Naive Bayes

Lasso

- glmnet
- cv.glmnet
- Upweighting

Useful ~ Negative

Linear regression
`lm(log_useful ~
binary_negative)`

Find the proportion
of above avg. useful
reviews for negative
and positive

Results

Hand Coding

```

      0  1
0 70  2
1  3 25
Krippendorff's alpha

Subjects = 100
Raters   = 2
alpha    = 0.875

```

Coder_1

28% negative

Coder_2

27% negative

ML Models

Lasso

- glmnet
- cv.glmnet
- Upweighting

Naive Bayes (best)

```

Precision : 0.5556
Recall    : 0.8333
F1        : 0.6667

```

Useful ~ Negative

Linear regression

p-value of 0.854

Adj R2 of 0.0004

t.negative <dbl>	useful_prop <dbl>
0	0.5718964
1	0.6139911

Limitations

Making a binary out of sentiment and useful lead to oversimplification.

NLP struggles with context, sarcasm, and informal language.

Next Steps

Alternative Prediction

- Trees
- Random Forest
- Gemini

Control/Consider Other Variables

Continuous Sentiment Score with a Quanteda Sentiment Dictionary